KPI Definition Sheet

AnalystLab Africa Experience Lab | Week 3 | Track A: Data Analytics | Author: Divine Okezie Nwaoha | Tools: Power BI, Power Query, DAX, Python/SQL (validation)

All KPIs are DAX measures. Table and column names are as loaded in Power BI.

KPI	Definition	DAX	Notes
Total Encounters	Count of encounter records	`COUNTROWS(Encounters)`	Benchmark 27,891
Unique Patients	Distinct patients with at least one encounter	`DISTINCTCOUNT(Encounters[PATIENT])`	Benchmark 974
Total Claim Cost	Sum of total claim cost	`SUM(Encounters[TOTAL_CLAIM_COST])`	About $101.5M
Avg Claim Cost	Total claim cost per encounter	`DIVIDE([Total Claim Cost],[Total Encounters])`	Includes 89 zero-cost claims
Total Payer Coverage	Sum of payer-covered amount	`SUM(Encounters[PAYER_COVERAGE])`	
Coverage Rate	Share of claim cost covered by payers	`DIVIDE([Total Payer Coverage],[Total Claim Cost])`	About 30.6% overall
Zero Coverage Encounters	Encounters with coverage = 0	`CALCULATE([Total Encounters],Encounters[PAYER_COVERAGE]=0)`	Benchmark 13,586
Zero Coverage %	Zero-coverage encounters / all encounters	`DIVIDE([Zero Coverage Encounters],[Total Encounters])`	48.7%
Zero Coverage Cost Share	Claim cost in zero-coverage encounters / total claim cost	`DIVIDE(CALCULATE([Total Claim Cost],Encounters[PAYER_COVERAGE]=0),[Total Claim Cost])`	62.2%
Insured Zero Coverage	Zero-coverage encounters excluding NO_INSURANCE	`CALCULATE([Total Encounters],Encounters[PAYER_COVERAGE]=0,Payers[Payer_Name]<>"NO_INSURANCE")`	About 4,779
Over 24h %	Encounters longer than 24h / all encounters	`DIVIDE(CALCULATE([Total Encounters],Encounters[Over_24h]=1),[Total Encounters])`	Duration = STOP minus START
Class Share %	Class encounters / all encounters	`DIVIDE([Total Encounters],CALCULATE([Total Encounters],ALL(Encounters[ENCOUNTERCLASS])))`	
Cost Share %	Class claim cost / total claim cost	`DIVIDE([Total Claim Cost],CALCULATE([Total Claim Cost],ALL(Encounters[ENCOUNTERCLASS])))`	
Procedure Count	Count of procedure records	`COUNTROWS(Procedures)`	
Avg Procedure Cost	Mean base cost	`AVERAGE(Procedures[BASE_COST])`	Skewed by rare high-cost items
Median Procedure Cost	Median base cost	`MEDIAN(Procedures[BASE_COST])`	
Avg Cost (Filtered)	Average cost shown only above a minimum volume	`IF([Procedure Count]>=[Min Procedure Volume Value],[Avg Procedure Cost])`	What-if parameter, default 30
Avg Encounters per Patient	Encounters / unique patients	`DIVIDE([Total Encounters],[Unique Patients])`	About 28.6 (median 14)

Transformations (Power Query), with reasons

- Dates converted to datetime; added `Duration_Hours`, `Over_24h`, `Zero_Coverage`, `Encounter_Date`, `Encounter_Year` (needed for duration, trend and coverage KPIs).

- Cost columns set to Fixed decimal; `CODE` columns set to Text (identifiers, not numbers).
- Patient names, addresses, birthplace, coordinates and zip removed (privacy; not needed).
- No duplicates removed and no missing values replaced (see the Data-Quality Assessment).